GMAT Data Insights

Multi-Source Reasoning

Training Exercises

Practice exercises covering Multi-Source Reasoning (MSR) question types. Complete after DI-05 lesson.

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

1. Training Overview

Session Details:

- **Total Questions:** 6 MSR sets (approximately 24 individual evaluations)
- **Recommended Time:** 25-30 minutes
- **Topics Covered:**
  - Synthesizing information from multiple sources
  - Cross-referencing data between tabs
  - Yes/No inference questions
  - Multiple-choice synthesis questions
  - Budget and constraint analysis
- **Complete After:** DI-05 (Multi-Source Reasoning) lesson

2. Multi-Source Reasoning Questions

MSR Question Format Reminder:

- Each set presents 2-3 tabs of information (text, tables, or both)
- You must navigate between tabs to find relevant information
- Questions may require combining data from multiple sources
- Question types include Yes/No, True/False, and multiple-choice
- Target time: 3-4 minutes per set

2.1. Set 1: Altered Designs Inventory Decision

Source: MSR-TRAINING-001

Tab 1 — Inventory Considerations

Altered Designs, a boutique furniture company, must select inventory for the next fiscal year. Five stakeholders will influence purchasing decisions for the primary showroom and warehouse. Each stakeholder rates four variables based on priority, with the CEO serving as tiebreaker if no majority emerges among the five decision-makers.

The inventory priority is determined by majority vote among the five executives. If three or more executives give a variable their highest priority rating (A), that variable becomes the top inventory priority. If no variable receives a majority of A ratings, the CEO's priorities determine the outcome.

Tab 2 — Decision Process

Five executives (CEO, CFO, CMO, Production Director, Design Director) rate four variables on a scale from A to D, where A represents highest priority and D represents lowest priority:

- **Cost:** Overall expense of acquiring inventory
- **Delivery time:** Speed of receiving inventory from suppliers
- **Demand:** Current market demand for product types
- **Durability:** Long-term quality and longevity of items

Tab 3 — Inventory Preferences

Executive	Cost	Delivery Time	Demand	Durability
CEO	B	C	A	D
CFO	A	D	B	C
CMO	D	B	A	C
Production Director	C	D	B	A
Design Director	C	B	D	A

For each of the following statements, select **Yes** if the statement can be inferred from the information provided, or **No** if it cannot.

Statement	Yes	No
Demand is currently the highest priority in selecting next season's inventory at Altered Designs.	○	○
If the CEO changes her top inventory priority to Cost, the overall top inventory priority would change to Cost.	○	○

The Production Director and Design Director are most aligned in their inventory priorities among the five stakeholders.	○	○

2.2. Set 2: Greenhouse Gas Emissions Analysis

Source: MSR-TRAINING-002

Tab 1 — Greenhouse Gas Emissions Overview

The global warming potential (GWP) of a greenhouse gas is measured in teragrams (Tg) of carbon dioxide equivalent (CO_2 Eq). In 2010, total U.S. greenhouse gas emissions were 6,821.8 Tg CO_2 Eq.

Carbon dioxide (CO_2) represented approximately 83.6% of total U.S. greenhouse gas emissions in 2010. Of all CO_2 emissions in the United States, fossil fuel combustion accounted for 94.4%.

Global fossil fuel emissions in 2010 totaled approximately 30,313 Tg CO_2. The United States contributed approximately 18% of these global emissions.

Tab 2 — U.S. CO_2 Emissions by Sector (2010)

The following table shows CO_2 emissions from fossil fuel combustion in the United States, broken down by sector:

Sector	Emissions (Tg CO_2)
Transportation	1,750.0
Industrial	827.4
Residential	1,183.7
Commercial	997.1
Total from Fossil Fuel Combustion	**5,380.3**

Note: Numbers may not sum exactly due to independent rounding.

For each of the following statements, select **Yes** if it can be inferred from the information provided, or **No** if it cannot.

Statement	Yes	No
In 2010, total U.S. CO_2 emissions were approximately 5,700 Tg.	○	○
The U.S. Transportation sector's CO_2 emissions exceeded 5% of global fossil fuel emissions in 2010.	○	○
In 2010, methane emissions in the U.S. were less than 6% of total greenhouse gas emissions.	○	○

Multiple Choice Question:

In 2010, CO_2 emissions from fossil fuel combustion in the United States accounted for approximately what percent of total greenhouse gas emissions in the United States?

○ 68.9% ○ 74.2% ○ 78.9% ○ 83.6% ○ 94.4%

2.3. Set 3: Island Museum Kaxna Collection — Part A (Dating Techniques)

Source: DI-GMAT-SK-00013

Tab 1 — Analysis Techniques

Island Museum analyzes historical artifacts using one or more techniques described below —all but one of which is performed by an outside laboratory—to obtain specific information about an object's creation. For each type of material listed, the museum uses only the technique described:

- **Animal teeth or bones:** The museum performs isotope ratio mass spectrometry (IRMS) in-house to determine the ratios of chemical elements present, yielding clues as to the animal's diet and the minerals in its water supply.
- **Metallic ores or alloys:** Inductively coupled plasma mass spectrometry (ICP-MS) is used to determine the ratios of traces of metallic isotopes present, which differ according to where the sample was obtained.
- **Plant matter:** While they are living, plants absorb carbon-14, which decays at a predictable rate after death; thus radiocarbon dating is used to estimate a plant's date of death.
- **Fired-clay objects:** Thermoluminescence (TL) dating is used to provide an estimate of the time since clay was fired to create the object.

Tab 2 — Kaxna Collection Background

Island Museum has acquired a collection of metal, fired clay, stone, bone, and wooden artifacts found on the Kaxna Islands, and presumed to be from the Kaxna Kingdom of 1250-850 BC. Researchers have mapped all the mines, quarries, and sources of clay on Kaxna and know that wooden artifacts of that time were generally created within 2 years after tree harvest. There is, however, considerable uncertainty as to whether these artifacts were actually created on Kaxna.

In analyzing these artifacts, the museum assumes that radiocarbon dating is accurate to approximately ±200 years and TL dating is accurate to approximately ±100 years.

Tab 3 — Budget Constraints

For outside laboratory tests, the museum's first-year budget for the Kaxna collection allows unlimited IRMS testing, and a total of $7,000—equal to the cost of 4 TL tests plus 15 radiocarbon tests, or the cost of 40 ICP-MS tests—for all other tests. For each technique applied by an outside lab, the museum is charged a fixed price per artifact.

Question 3A: For each artifact type below, indicate whether a range of dates for the object's creation can be obtained using one of the techniques described.

Artifact	Yes	No
Bronze statue of a deer	○	○

Fired-clay pot	○	○
Wooden statue of a warrior	○	○

2.4. Set 3: Island Museum Kaxna Collection — Part B (Kingdom Confirmation)

Source: DI-GMAT-SK-00014

Use the same tabs from Set 3 Part A.

Question 3B: For each artifact and analysis result below, indicate whether the result confirms the artifact was created during the time of the Kaxna Kingdom (1250-850 BC).

Artifact and Result	Yes	No
Bone necklace shown by IRMS to have element ratios characteristic of artifacts known to be from the Kaxna Kingdom	○	○
Fired-clay jug dated to 1050 BC by TL dating	○	○
Copper box shown by ICP-MS to have the same ratio of trace metals found in the copper mines of Kaxna	○	○

2.5. Set 3: Island Museum Kaxna Collection — Part C (Budget Analysis I)

Source: DI-GMAT-SK-00015

Use the same tabs from Set 3 Part A.

Question 3C: For each collection of artifacts below, indicate whether the cost of all pertinent analysis techniques can be shown to be within the museum's first-year Kaxna budget.

Artifact Collection	Yes	No
2 fired-clay statues and 10 bronze statues	○	○
3 fired-clay statues and 5 tin implements	○	○
4 fired-clay pots and 20 wooden statues	○	○

2.6. Set 3: Island Museum Kaxna Collection — Part D (Budget Analysis II)

Source: DI-GMAT-SK-00016

Use the same tabs from Set 3 Part A.

Question 3D: For each collection of artifacts below, indicate whether the cost of all pertinent analysis techniques can be shown to be within the museum's first-year Kaxna budget.

Artifact Collection	Yes	No
2 bone implements and 5 fired-clay cups decorated with gold	○	○
7 wooden statues and 20 metal implements	○	○
15 wooden statues decorated with bone	○	○

2.7. Set 3: Island Museum Kaxna Collection — Part E (Multiple Choice)

Source: DI-GMAT-SK-00017 & DI-GMAT-SK-00018

Use the same tabs from Set 3 Part A.

Question 3E: Among the Kaxna artifacts is a wooden box containing both a small fired-clay bead and some river sediment containing clay and plant matter. Based on the museum's assumptions, which one of the following details about the bead can be determined by applying one of the tests in the manner described?

○ A range of dates for its manufacture

○ The Kaxna island on which it was made

○ Vegetation patterns near the workshop where it was made

○ A range of dates for its placement in the box

○ The source of clay used to make the bead

Question 3F: Which one of the following pieces of information would, on its own, provide the strongest evidence that a given artifact was actually produced on Kaxna?

○ A radiocarbon date of 1050 BC for a wooden bowl

○ IRMS analysis of a necklace made from animal bones and teeth

○ A TL date for a fired-clay brick that places it definitively in the period of the Kaxna Kingdom

○ ICP-MS analysis of a metal tool that reveals element ratios unique to a mine on Kaxna

○ Determination that a stone statue was found near a quarry known to produce stone statues during the Kaxna Kingdom

2.8. Set 4: Marketing Survey Project

Source: MSR-TRAINING-004

Tab 1 — Email from Project Administrator

From: Sarah Chen, Project Administrator
To: Research Team
Subject: Marketing Survey Response Rate
Date: March 15

Team,

As you know, we need a statistically significant sample size for our consumer behavior study. Our target is 700 completed surveys. Our budget allows us to compensate survey respondents at $15 per completed survey, and we have allocated $12,000 total for respondent compensation.

Please let me know the current status of survey responses. If we haven't reached our target, we may need to extend the survey period or expand our outreach methods.

Best, Sarah

Tab 2 — Email from Project Coordinator

From: Marcus Webb, Project Coordinator
To: Sarah Chen
Subject: RE: Marketing Survey Response Rate
Date: March 16

Sarah,

As of this morning, we have received 350 completed surveys. Based on current response rates, we project receiving approximately 50 additional responses per day if we maintain our current outreach efforts.

We still have 10 days until our planned survey close date. However, I should note that response rates typically decline in the final days of a survey period.

Should I prepare contingency plans for additional outreach?

Marcus

Tab 3 — Budget Summary

Item	Amount
Total Budget Allocated	$15,000
Survey Platform Fees	$2,000
Respondent Compensation	$12,000
Contingency Fund	$1,000

Note: We have agreed to try not to exceed the allocated budget. Any expenses beyond the total budget would require additional approval from the department head.

For each of the following statements, select **Yes** if it can be inferred from the information provided, or **No** if it cannot.

Statement	Yes	No
The survey has currently received enough responses to meet the target sample size.	○	○
Based on projected response rates, the survey will definitely reach its target of 700 responses before the close date.	○	○
It is possible that the project will need to exceed the allocated respondent compensation budget to reach the target sample size.	○	○